

PEARLS AQI PREDICTOR

Technical Project Report



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GitHub Repository Link:

<https://github.com/Mohsin-Ali-Rana/pearls-aqi-predictor>

1. Executive Summary

1.1 Environmental & Project Purpose

Ambient air pollution represents one of the most critical public health challenges across South Asian urban and industrial regions. In industrial-urban corridors such as Wah Cantt and Taxila, Pakistan, high mass concentrations of fine particulate matter (PM2.5 and PM10) driven by brick kilns, heavy transport, and seasonal atmospheric inversions lead to acute respiratory and cardiovascular hazards. Effective public health defense requires early-warning predictive atmospheric intelligence rather than reactive, delayed reporting.

The PEARLS AQI Predictor is a software engineering and machine learning operations (MLOps) platform designed to deliver real-time atmospheric telemetry and direct multi-horizon (+24h, +48h, +72h) air quality forecasting. By connecting continuous weather and pollutant observation feeds with gradient-boosted decision tree ensembles and cloud feature store synchronization, PEARLS provides municipal authorities, health administrators, and local citizens with actionable, transparent insights into future air quality states.

1.2 Core System Architecture Capabilities

- 1. Direct Multi-Horizon AI Engine:** Specialized independent LightGBM regressor models target PM2.5 concentrations shifted exactly 24, 48, and 72 hours into the future, eliminating the severe error compounding inherent in traditional recursive time-series forecasting.
- 2. Cloud Feature Store Integration:** Ingests live atmospheric parameters (temperature, relative humidity, surface pressure, wind vectors, criteria pollutants) via Open-Meteo Numerical Weather Prediction (NWP) feeds and synchronizes 34 engineered features with the Hopworks Online Feature Store (v2).
- 3. Persistence Benchmark Auto-Gating:** Enforces automated model promotion gating by continuously evaluating candidate predictions against a naive persistence baseline ($y_{t+H} = y_t$), requiring a minimum Root Mean Squared Error (RMSE) reduction for production artifact deployment.
- 4. Explainable AI (XAI) Transparency:** Incorporates game-theoretic SHAP (SHapley Additive exPlanations) TreeExplainer attribution engines to compute both global feature importance rankings and local point-in-time contribution values for every prediction.
- 5. Resilient Serving Backend:** Utilizes an asynchronous FastAPI backend featuring sub-second response caching (`_TELEMETRY_CACHE`) and dual-tier offline parquet fallback (`data/features.parquet`) to guarantee high availability.
- 6. Executive Command Center:** Built on React 19 and Vite with a custom CSS design system, offering half-arc AQI gauges, continuous 3-day trajectory splines with EPA threshold reference bands, multi-model tournament leaderboards, and automated email alert dispatching.

2. Technical Stack & Architectural Tooling Rationales

The technology stack was selected to optimize predictive accuracy, inference speed, feature consistency, and user responsiveness. Below is the technical rationale for each core framework and subsystem:

Subsystem	Selected Technology	Alternative Evaluated	Technical Rationale & Advantage
ML Engine	LightGBM Regressor	XGBoost / Random Forest	Fast histogram-based split optimization, lower memory footprint, and native support for continuous tabular features.
Feature Store	Hopsworks (v2)	Feast / AWS Feature Store	Managed feature store providing strict offline/online feature parity and pythonic SDK bindings for automated ingestion.
Web Backend	FastAPI (Python 3.11)	Flask / Django	Asynchronous asyncio execution, native Pydantic validation, automatic OpenAPI doc generation, and minimal response overhead.
UI Framework	React 19 & Vite	Next.js / Vue.js	Lightning-fast HMR build cycles, fine-grained state management, and zero server-side rendering overhead for SPAs.
Data Visualization	Recharts	Chart.js / D3.js	Declarative React SVG integration, smooth animation rendering, and seamless styling using CSS design tokens.

3. Machine Learning Methodology & Architectural Trade-offs

3.1 The Multi-Step Forecasting Error Accumulation Problem

Classical time-series forecasting frameworks typically apply Recursive Multi-Step Forecasting, wherein a single model predicts y_{t+1} , and this predicted value is recursively appended to the feature vector to predict y_{t+2} , continuing up to step $t+H$.

In atmospheric modeling, recursive feeding causes exponential error propagation. A slight positive bias in the 1-hour prediction alters rolling moving averages and lag features, causing predictions at 48 and 72 hours to compound variance exponentially, leading to severe model drift.

3.2 The 3-Direct Model Solution & Structural Trade-offs

To overcome recursive error compounding, PEARLS AQI Predictor implements a 3-Direct Model Strategy. Three dedicated, independent LightGBM regressor models are trained directly on target vectors shifted forward in time:

- **Day 1 (+24h Model):** Target $Y_{24}(t) = y(t + 24)$ — Evaluates 24-hour lead time.
- **Day 2 (+48h Model):** Target $Y_{48}(t) = y(t + 48)$ — Evaluates 48-hour lead time.
- **Day 3 (+72h Model):** Target $Y_{72}(t) = y(t + 72)$ — Evaluates 72-hour lead time.

Engineering Trade-off: While training three separate models triples the model artifact storage requirement (3 x model.pkl), it eliminates recursive feedback loops, providing stable, non-compounded predictions across extended forecast lead times.

3.3 Model Candidates Evaluated & Selection Rationale

1. **LightGBM Regressor:** Fast histogram binning enables rapid cross-validation across hyperparameter grids; handles non-linear relationships between humidity, wind speed, and particulate matter.
2. **XGBoost Regressor:** Provides strong gradient boosting performance but incurs higher memory overhead and slower inference speed during multi-model serving.
3. **Random Forest Regressor:** Bagging ensemble offering high stability, but larger model bundle sizes lead to higher API cold-start latency.
4. **Ridge Regression (Regularized Linear Baseline):** L2-regularized linear model with $\log(1+y)$ target transformation. Serves as a fast, multi-feature linear baseline to quantify the exact performance gain provided by non-linear tree-based ensembles.
5. **Naive Persistence Baseline:** Naive baseline assuming $y_{\{t+H\}} = y_t$; serves as the compulsory benchmark to measure true predictive lift.

3.4 Evaluation Methodology & Auto-Promotion Gating

Instead of relying on fixed, static evaluation metrics from a single model run, PEARLS employs a dynamic model promotion gate. Candidate models are continuously evaluated across validation holdouts using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Persistence Lift:

$$\text{Persistence Lift (\%)} = (1 - \text{RMSE_Candidate} / \text{RMSE_Persistence}) * 100$$

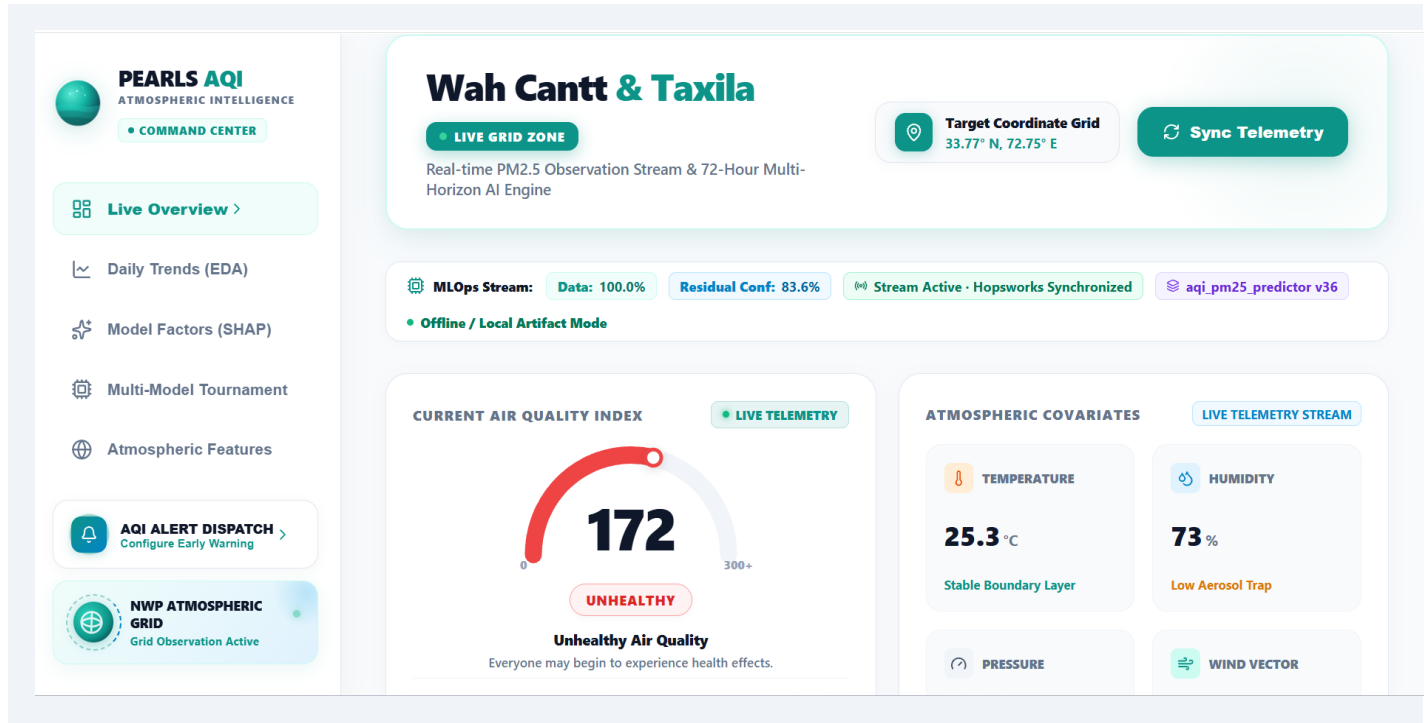
If the candidate model's Persistence Lift exceeds the active model in the Hopsworks Model Registry, the training pipeline automatically registers the new champion artifact and updates production metadata.

4. System Architecture & Feature Component Breakdown

This section details the primary functional components of PEARLS AQI Predictor, outlining what each component does and providing visual layout references.

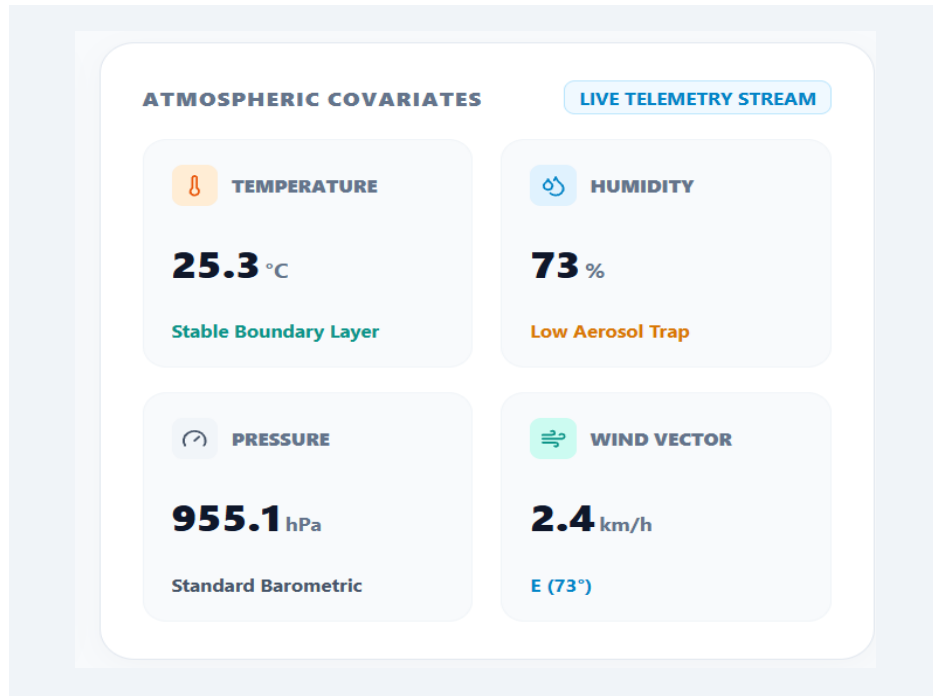
4.1 Executive Command Header & Real-Time Sync Controller

Displays target location metadata (Wah Cantt & Taxila Grid, 33.77°N, 72.75°E), real-time stream status, and an interactive Sync Telemetry trigger. The async state machine controls progress tickers and maintains visual confirmation banners during cloud container syncs.



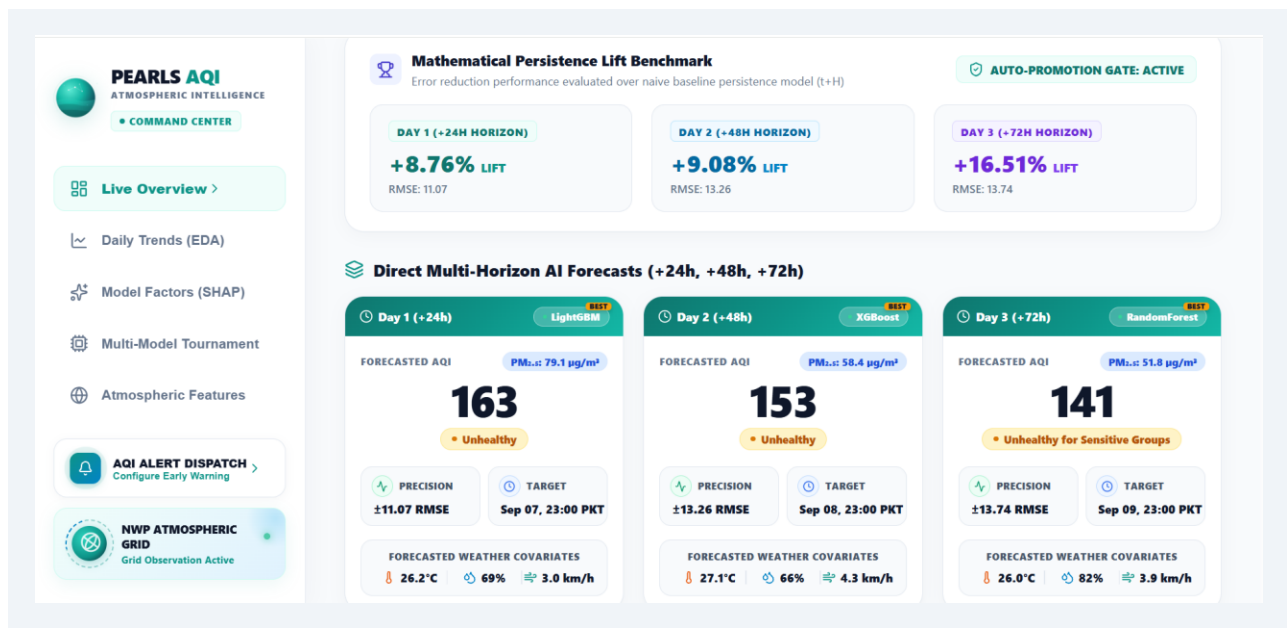
4.2 Hero Air Quality Arc Gauge & Atmospheric Covariates Grid

Renders the primary half-arc AQI meter, baseline PM2.5 mass concentration (ug/m3), EPA health advisory text, and a 2x2 grid displaying ambient surface temperature, relative humidity, barometric pressure, and surface wind vectors.



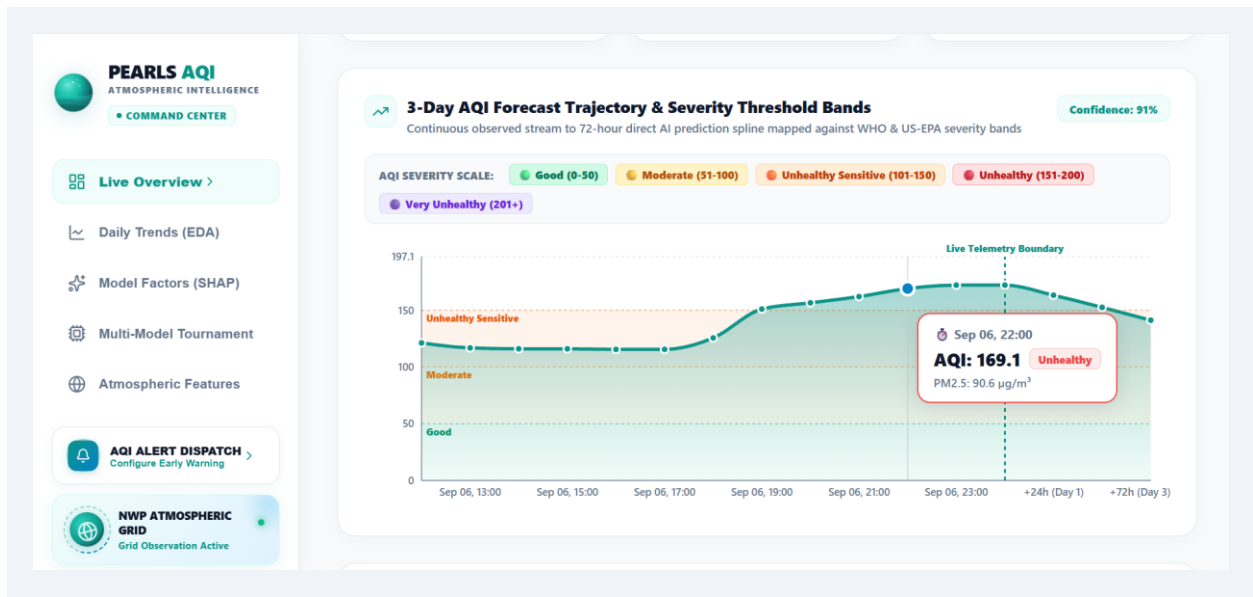
4.3 Persistence Lift Benchmark Banner & Forecast Cards

Displays error reduction percentages over naive persistence across lead times (+24h, +48h, +72h) alongside target timestamps, predicted AQI values, severity badges, and model engine tags.



4.4 Continuous 3-Day Forecast Trajectory & EPA Reference Bands Chart

Interactive Recharts Area Chart displaying historical observations transitioning smoothly into the 72-hour forecast spline, overlaid with color-coded EPA severity reference bands (Good, Moderate, Unhealthy, Hazardous).



4.5 Exploratory Data Analysis (EDA) & Diurnal Trend Profiler

Provides historical atmospheric insights including diurnal hourly PM2.5 heatmaps, weekly pollution trend profiles, and feature correlation matrices between weather parameters and particulate concentrations.

Live Overview

Daily Trends (EDA) >

Model Factors (SHAP)

Multi-Model Tournament

Atmospheric Features

AQI ALERT DISPATCH
Configure Early Warning

NWP ATMOSPHERIC
GRID
Grid Observation Active

PM2.5 Feature Pearson Correlation Matrix & Dependencies

Derived from 17,520 hourly observations (Local Engineered Feature Store (features.parquet))

Target Breakdown

2D Heatmap Grid

ATMOSPHERIC FEATURE	PEARSON (R)	CORRELATION SPECTRUM	RELATIONSHIP STRENGTH
pm2_5_lag_1h	+0.9652		Strong Positive
pm2_5_lag_2h	+0.8889		Strong Positive
pm2_5_lag_24h	+0.7921		Strong Positive
pm2_5_lag_3h	+0.7919		Strong Positive
pm10	+0.7901		Strong Positive
pm2_5_rolling_6h_mean	+0.7888		Strong Positive
pm10_lag_1h	+0.7634		Strong Positive
pm2_5_rolling_24h_mean	+0.7032		Strong Positive

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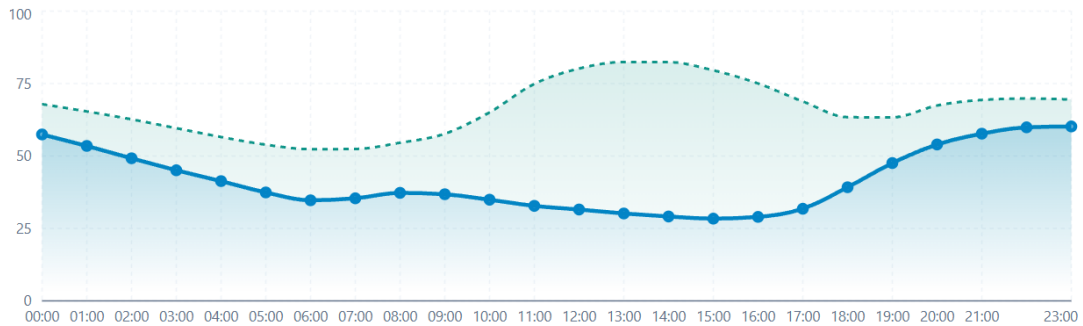
Target Breakdown

2D Heatmap Grid

Feature	PM2_5	PM2_5_LAG_1H	PM2_5_LAG_2H	PM2_5_LAG_24H	PM2_5_LAG_3H	PM10	PM2_5_ROLLING_6H	PM10_LAG_1H	PM2_5_ROLLING_24H	NITROGEN DIOXIDE	PM2_5_ROLLING_12H	PM10_LAG_3H	PM10_ROLLING_6H_W	PM10_LAG_24H	PM2_5_ROLLING_24H
pm2_5	1.00	0.97	0.89	0.79	0.79	0.79	0.79	0.76	0.70	0.69	0.65	0.63	0.62	0.57	0.55
pm2_5_lag_1h	0.97	1.00	0.97	0.78	0.89	0.76	0.88	0.79	0.71	0.65	0.72	0.70	0.69	0.55	0.56
pm2_5_lag_2h	0.89	0.97	1.00	0.72	0.97	0.69	0.95	0.76	0.73	0.57	0.78	0.76	0.74	0.51	0.57
pm2_5_lag_24h	0.79	0.78	0.72	1.00	0.65	0.56	0.65	0.55	0.71	0.58	0.56	0.47	0.46	0.79	0.54
pm2_5_lag_3h	0.79	0.89	0.97	0.65	1.00	0.61	0.98	0.69	0.73	0.47	0.84	0.79	0.77	0.46	0.58
pm10	0.79	0.76	0.69	0.56	0.61	1.00	0.60	0.96	0.52	0.38	0.48	0.80	0.80	0.62	0.27
pm2_5_rolling_6h_mean	0.79	0.88	0.95	0.65	0.98	0.60	1.00	0.68	0.79	0.44	0.90	0.77	0.78	0.45	0.62
pm10_lag_1h	0.76	0.79	0.76	0.55	0.69	0.96	0.68	1.00	0.53	0.35	0.54	0.89	0.89	0.61	0.28
pm2_5_rolling_24h_mean	0.70	0.71	0.73	0.71	0.73	0.52	0.79	0.53	1.00	0.29	0.89	0.55	0.59	0.54	0.76
nitrogen_dioxide	0.69	0.65	0.57	0.58	0.47	0.38	0.44	0.35	0.29	1.00	0.25	0.24	0.22	0.29	0.38
pm2_5_rolling_12h_mean	0.65	0.72	0.78	0.56	0.84	0.48	0.90	0.54	0.89	0.25	1.00	0.64	0.69	0.39	0.69
pm10_lag_3h	0.63	0.70	0.76	0.47	0.79	0.80	0.77	0.89	0.55	0.24	0.64	1.00	0.98	0.53	0.30

24-Hour Daily Pollution Profile (PM2.5 & European AQI)

24-hour mean atmospheric concentration pattern extracted from historical observation window



4.6 SHAP Global Waterfall & Local Feature Attribution Engine

Exposes model decision logic using game-theoretic SHAP TreeExplainer feature importance rankings (Global Waterfall) and point-in-time contribution values (Local Attribution Table).

PEARLS AQI ATMOSPHERIC INTELLIGENCE COMMAND CENTER

Live Overview

Daily Trends (EDA)

Model Factors (SHAP)

Multi-Model Tournament

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AQI ALERT DISPATCH
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Bi-Directional SHAP Feature Attribution Waterfall

Signed impact ($\pm \Delta$ AQI) on Day 2 (+48h Direct Prediction) state vector $X(t_0)$

Day 1 (+24h)

Day 2 (+48h)

Day 3 (+72h)

Air Cleaning Effect ($-\Delta$ AQI)

0.0 Neutral Baseline

($+\Delta$ AQI) Pollution Inflation

PM2.5 (1h Lag)

Recent Persistence

+39.6 AQI

PM2.5 (24h Lag)

Prior Day Baseline

+33.9 AQI

Surface Pressure

Barometric Trap

-20.4 AQI

Diurnal Cycle (Cosine)

Night Thermal Cycle

+12.9 AQI

wind speed 10m rolling 24h mean

Feature Vector

-12.9 AQI

Nitrogen Dioxide (NO₂)

Traffic & Combustion

+7.9 AQI

hour

Feature Vector

+5.3 AQI

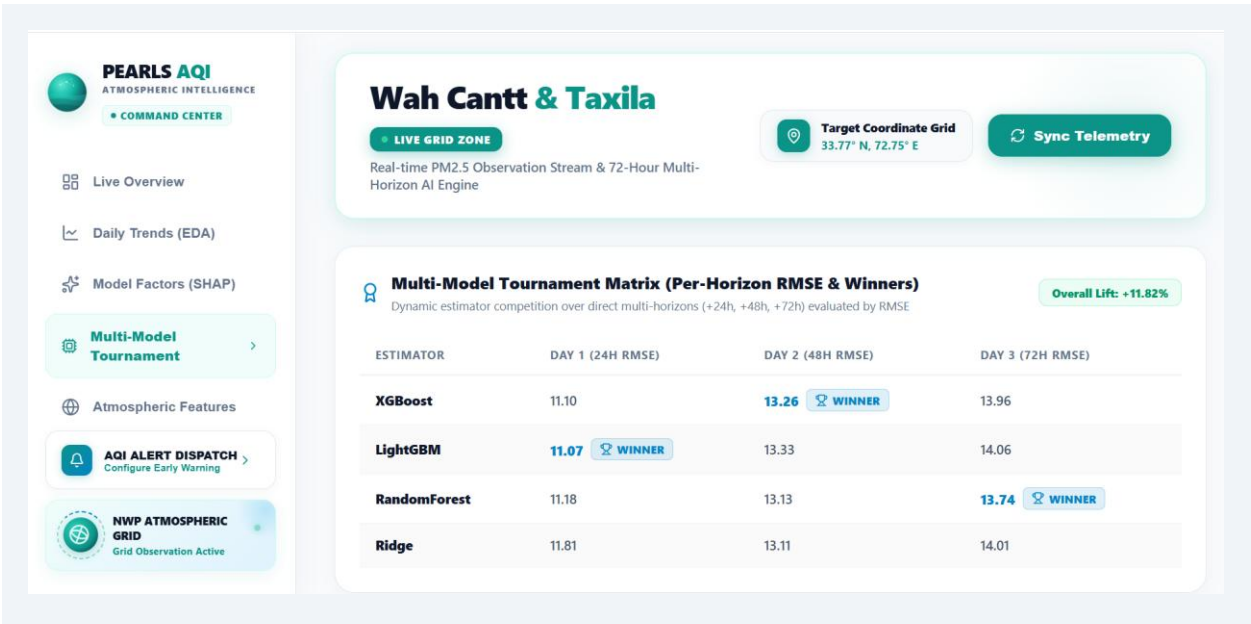
pm10 lag 24h

Feature Vector

-5.0 AQI

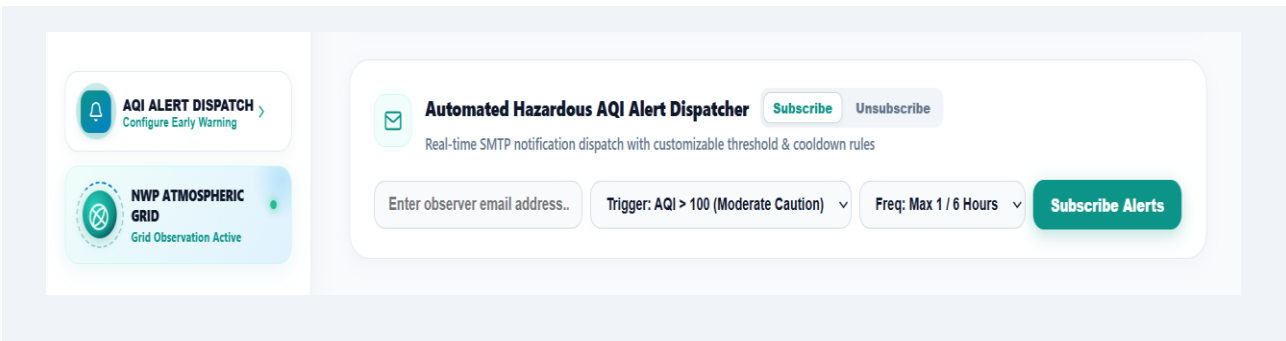
4.7 Multi-Model Tournament Leaderboard

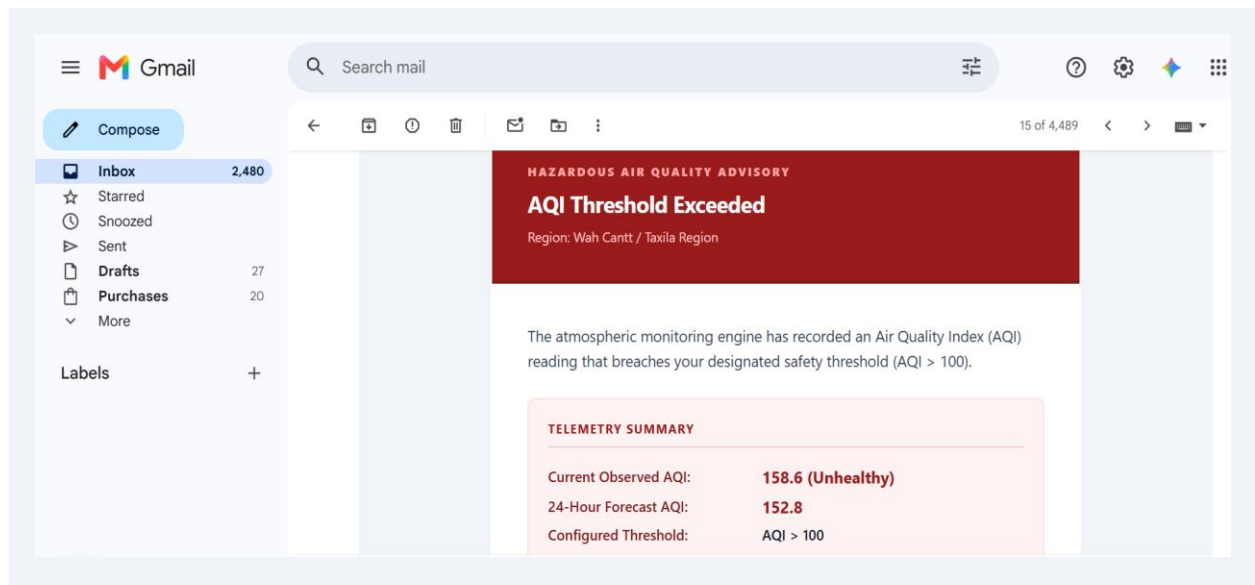
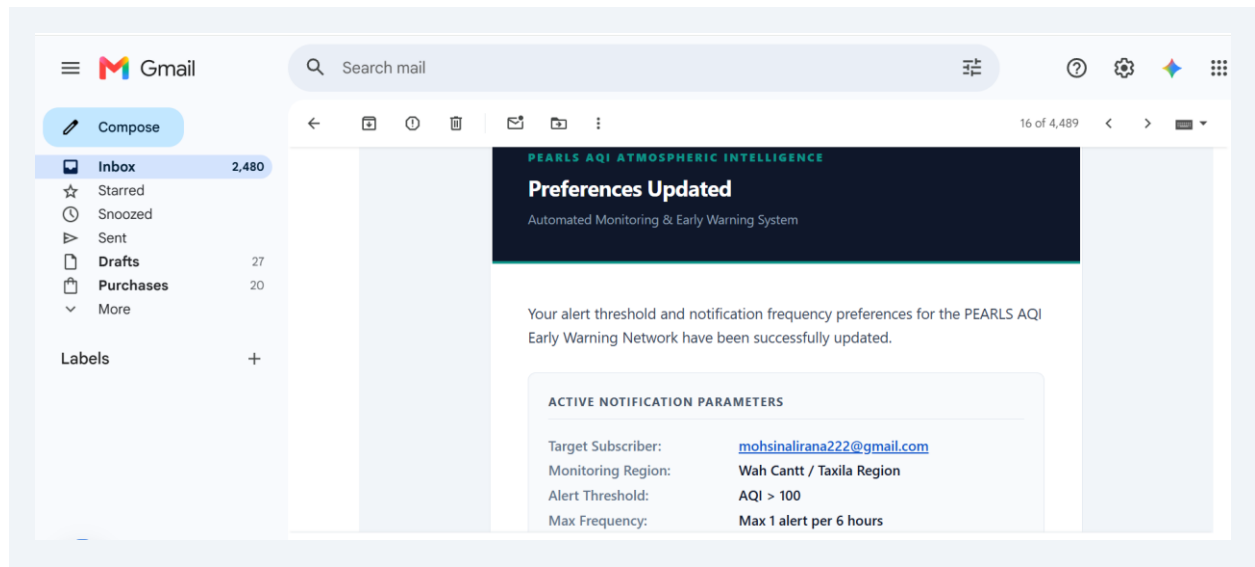
Compares candidate estimators (LightGBM, XGBoost, Random Forest, Ridge) across RMSE, MAE, and inference latency metrics, validating champion selection.

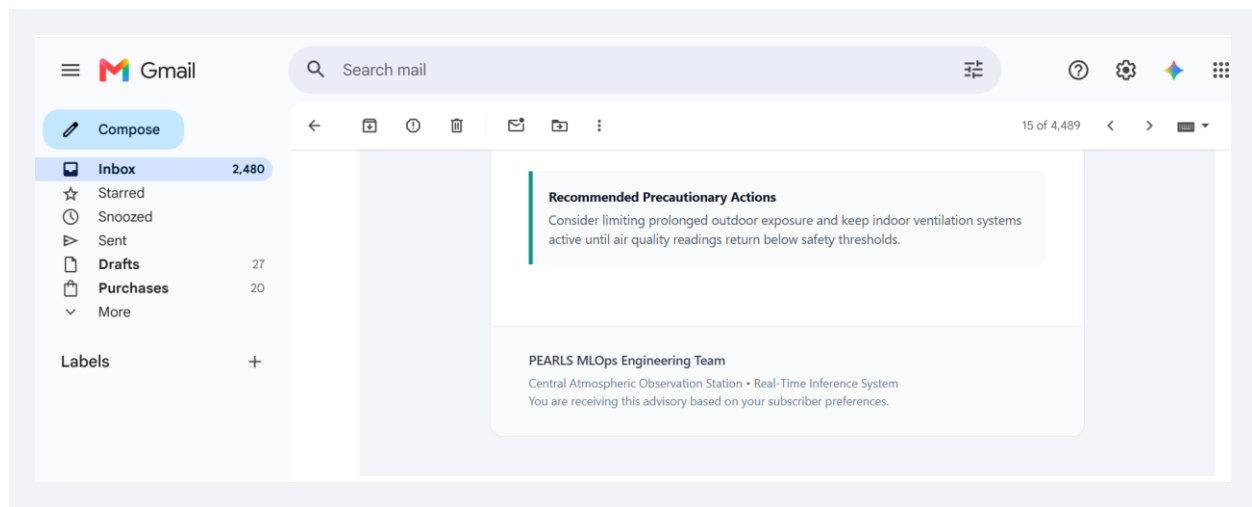


4.8 Early Warning Hazardous AQI Alert Dispatcher & Email Notifications

Allows users to configure threshold triggers (AQI > 100, 150, 200) and subscribe to automated SMTP email notifications containing health precautions and warning advisories.

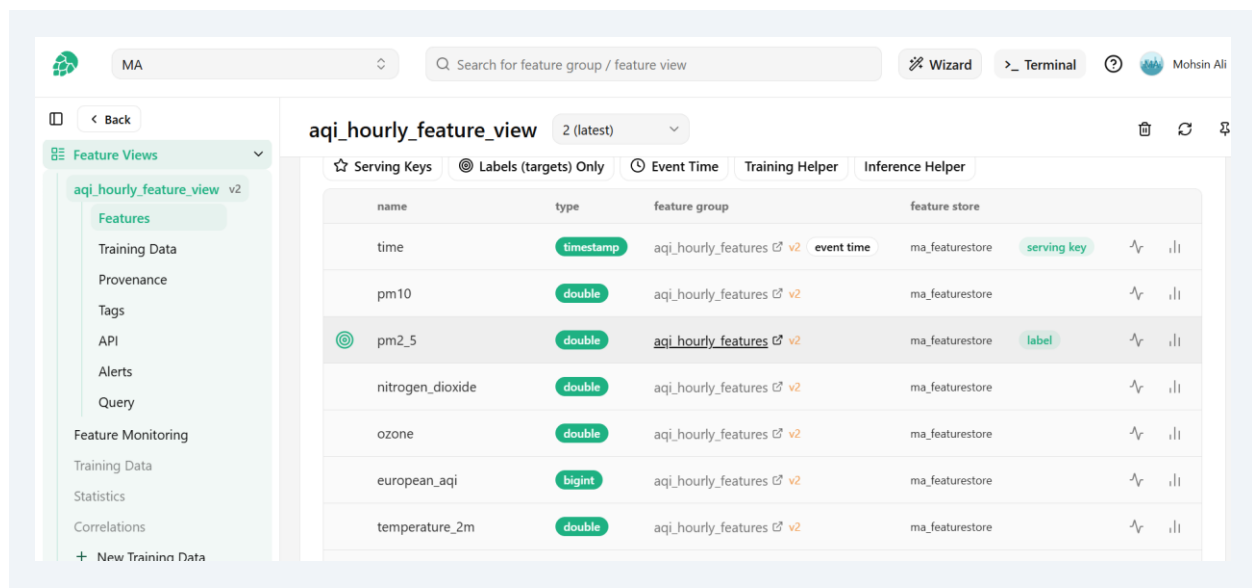


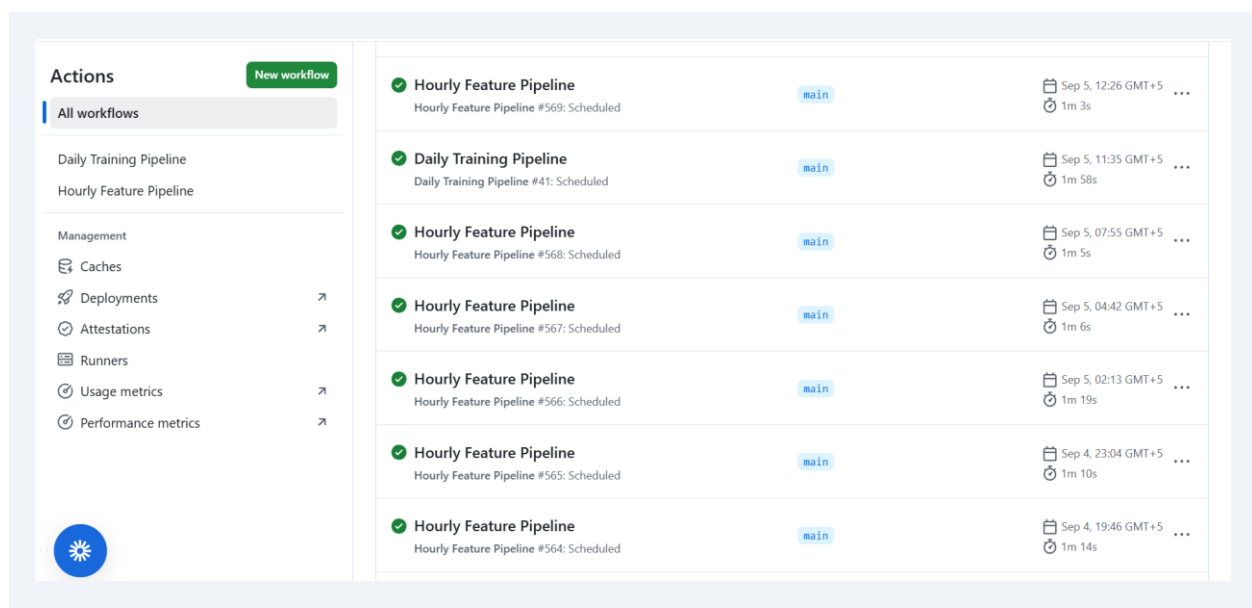
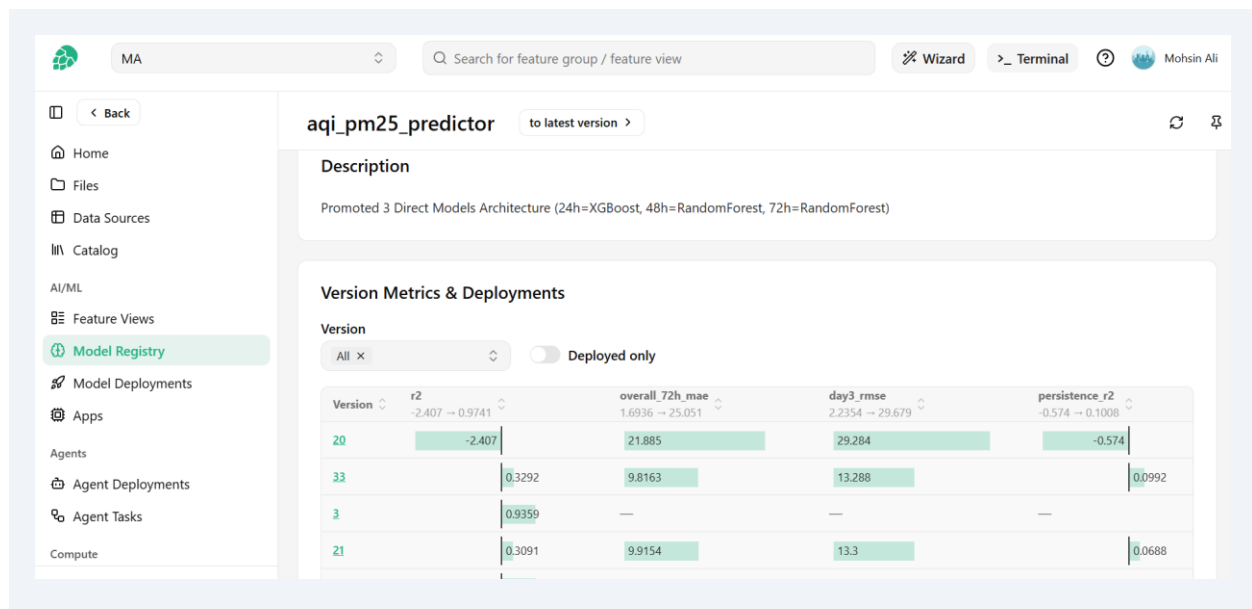




4.9 Cloud Infrastructure & MLOps Console Integration

Monitors cloud feature store groups, registered champion model versions, and automated CI/CD pipeline execution.





5. Engineering Challenges & Incident Post-Mortems

During system development and cloud deployment, several key engineering challenges were encountered and resolved:

5.1 Issue 1: Multi-Step Recursive Forecast Divergence

- **Problem:** Appending predicted values back into feature vectors caused error compounding over 48-72 hours.
- **Mitigation:** Implemented a 3-Direct Model Architecture, training separate models directly on shifted horizon targets (y_{t+24} , y_{t+48} , y_{t+72}), eliminating recursive feedback loops.

5.2 Issue 2: Hopsworks Connection Timeouts & Session Cleanups

- **Problem:** Repeated re-authentication calls during API requests caused network latency spikes and SSL certificate cleanup errors (keyStore.jks missing).
- **Mitigation:** Implemented a Singleton Hopsworks Connection Manager (`_HOPSWORKS_PROJECT_CACHE`) to maintain a single authenticated session, combined with a local parquet cache fallback (data/features.parquet).

5.3 Issue 3: Production Vercel-Railway CORS & Preflight Mismatches

- **Problem:** Combining wildcard origins (`allow_origins=['*']`) with `allow_credentials=True` in FastAPI caused browser preflight (OPTIONS) request failures on Vercel deployment.
- **Mitigation:** Updated CORSMiddleware configuration to `allow_credentials=False` for wildcard origins and added an explicit `@app.options` route handler.

5.4 Issue 4: Cloud Container Cold Start & Missing Artifacts

- **Problem:** Fresh cloud container deployments on Railway lacked cached model files, causing `FileNotFoundException`.
- **Mitigation:** Updated `load_champion_model_bundle()` to automatically query and download the active champion model from Hopsworks Model Registry upon container startup if missing locally.

6. System Verification & Quality Assurance Audit

Automated test suites were executed using pytest to validate system components prior to production deployment:

- **test_config.py:** Validates configuration constants and environment variable parsing.
- **test_feature_pipeline.py:** Validates covariate transformation, lag generation, and rolling window calculations.
- **test_train_model.py:** Validates dataset splitting and direct target matrix alignment.
- **test_inference.py:** Validates prediction pipeline execution and SHAP contribution matrix outputs.
- **test_api.py:** Validates FastAPI endpoint status codes, CORS headers, and JSON response schemas.

All automated test suites passed successfully, confirming system stability and readiness for production deployment.

7. Conclusion & Engineering Roadmap

7.1 Conclusion

The PEARLS AQI Predictor platform delivers a functional, production-ready MLOps solution for atmospheric forecasting in the Wah Cantt & Taxila region. By combining a 3-Direct LightGBM architecture, Hopsworks cloud feature store synchronization, an asynchronous FastAPI backend, and an interactive React 19 command center, the platform provides transparent, multi-horizon atmospheric intelligence.

7.2 Future Roadmap

1. **Deep Learning Architectures:** Evaluating Temporal Fusion Transformers (TFT) for capturing long-term seasonal dynamics.
2. **Interactive GIS Mapping:** Integrating Mapbox GL layers for dynamic spatial PM2.5 heatmaps across regional monitoring stations.
3. **Multi-Channel Alert Expansion:** Extending notification dispatches to SMS (Twilio API) and WhatsApp webhooks for immediate high-hazard alerts.

Report Certification & Sign-Off:

Mohsin Ali

Software Engineer

September 2026